Taking a Step Back

When I first encountered this idea many years ago, it was a bit of a shock. Up until that point, I had thought I understood the concept of entropy fairly well, but the fact of the matter was that, following the approach of textbooks I'd studied, I'd only ever considered entropy's implications for the future. And, as we've just seen, while entropy applied toward the future confirms our intuition and experience, entropy applied toward the past just as thoroughly contradicts them. It wasn't quite as bad as suddenly learning that you've been betrayed by a longtime friend, but for me, it was pretty close.

Nevertheless, sometimes it's good not to pass judgment too quickly, and entropy's apparent failure to live up to expectations provides a case in point. As you're probably thinking, the idea that all we're familiar with just popped into existence is as tantalizing as it is hard to swallow. And it's not "merely" that this explanation of the universe challenges the veracity of everything we hold to be real and important. It also leaves critical questions unanswered. For instance, the more ordered the universe is today— the greater the dip in Figure 6.4—the more surprising and unlikely is the statistical aberration required to bring it into existence. So if the universe could have cut any corners, making things look more or less like what we see right now while skimping on the actual amount of order, probabilistic reasoning leads us to believe it would have. But when we examine the universe, there seem to be numerous lost opportunities, since there are many things that are more ordered than they have to be. If Michael Jackson never recorded *Thriller* and the millions of copies of this album now distributed worldwide all got there as part of an aberrant fluctuation toward lower entropy, the aberration would have been far less extreme if only a million or a half-million or just a few albums had formed. If evolution never happened and we humans got here via an aberrant jump toward lower entropy, the aberration would have been far less extreme if there weren't such a consistent and ordered evolutionary fossil record. If the big bang never happened and the more than 100 billion galaxies we now see arose as an aberrant jump toward lower entropy, the aberration would have been less extreme if there were 50 billion, or 5,000, or just a handful, or just one galaxy. And so if the idea that our universe is a statistical fluctuation—a